# stent_capture

Magnetic capture of SPION-labelled cells
on a cerebral ferromagnetic stent

## Project Overview

Stages 1 – 3 complete

A four-stage Python framework for high-resolution
trajectory-level exploration of magnetic cell targeting,
complementing the dissertation COMSOL finite-element model.

v0.5.0 — 2026-04-11

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# 1 Introduction

This document walks the reader through the stent_capture project: a Python simulation framework that models the magnetic capture of SPION-labelled endothelial cells on a ferromagnetic stent deployed under cerebral flow conditions. It is intended as a standalone companion to the dissertation, sufficient for a supervisor or examiner to understand the physics, the code, and the numerical results without having to read the source tree.

What the project simulates. A ring of eight uniformly magnetised rectangular struts sits inside a cylindrical cerebral artery modelled with Poiseuille flow. A uniform external field B0 is superposed on the stent's analytical magnetostatic field. SPION-labelled cells are introduced upstream and their trajectories are integrated under a terminal-velocity ODE that balances Stokes drag against the magnetic dipole force. The framework evaluates both the static capture criterion |F_mag| > |F_drag| and the full trajectory integration, and compares them at MCA-representative flow and loading conditions.

Why the project exists. The dissertation proper uses a finite-element COMSOL model to solve the full coupled magneto-hydrodynamic problem. That model is expensive: each configuration costs minutes to hours, making dense parametric sweeps infeasible. The stent_capture framework is a supplementary tool built on fast analytical expressions (Akoun & Yonnet 1984) so that (a) parameter sweeps across loadings, velocities, and stent geometries run in seconds to minutes rather than hours; (b) trajectory ensembles of hundreds of cells can be integrated without meshing; and (c) the results serve as an independent numerical cross-check on the COMSOL simulations.

# 1 Introduction (cont.)

High-level architecture — the four stages. The project is organised as four sequential stages, each building on the previous one and each leaving behind its own tests and figures so that regressions surface immediately.

Stage 1 — Field model. Implements the Akoun & Yonnet (1984) closed-form expressions for the B-field of a uniformly magnetised rectangular prism, rotates and superposes 8 struts into a ring, and layers an optional uniform external field B0 on top. Stage 1 establishes the field primitives (field_at, B_magnitude, grad_B) that every later stage consumes through a shared TotalField interface.

Stage 2 — Static capture criterion. Adds the SPION cell model, the Furlani & Ng (2006) scalar-gradient magnetic force, the Poiseuille-flow Stokes drag, and the scalar capture criterion |F_mag| > |F_drag|. Produces 2D capture maps and SPION loading sweeps. Shows that the static criterion predicts no capture in the lumen at MCA flow for 10 pg loading — motivating Stage 3.

Stage 3 — Trajectory integration. Adds a low-Re terminal-velocity ODE solver built on scipy.integrate.solve_ivp, event-based capture detection, a multiprocessing capture-efficiency sweep, and the headline static-vs-trajectory comparison figure. Demonstrates that trajectory drift extends the effective capture range by ~14x over the static criterion at MCA conditions.

Stage 4 — Geometry optimisation (planned). Would sweep strut count, thickness, length, and magnetisation to identify optimal stent designs. Not in scope for this overview.

The rest of this document is organised to mirror that structure: Section 2 derives the physics; Section 3 describes the code that implements it; Section 4 walks through the key figures produced; Section 5 lists the known limitations; Section 6 explains how to run everything; and Section 7 collects the literature references.

# 2 Physical model

## 2.1 Geometry and coordinate frames

The stent is modelled as a ring of n = 8 identical rectangular struts equally spaced around a circle of radius R = 1.5 mm in the z = 0 plane. Each strut has radial thickness t = 80 um, circumferential width w = 100 um, and axial length L = 500 um. The ring is placed inside a straight cylindrical vessel of radius R_ves = 1.54 mm, so the stent outer surface is flush with the vessel wall and the inner lumen has radius r_inner = R - t/2 ~ 1.46 mm.

Two coordinate frames are used. The global frame (x, y, z) has the stent axis along +z and the ring centred at the origin. Each strut i has a local frame (x', y', z') aligned with its own radial, circumferential, and axial directions — this is the frame in which the Akoun-Yonnet kernel is evaluated. The global field is recovered by rotating the local field by the strut's azimuth theta_i = 2*pi*i / n and summing the 8 contributions.

Magnetisation direction is a user parameter (mag_mode = radial / circumferential / axial). All results in this document use the default mag_mode = 'radial', i.e. each strut is magnetised outward in its own local x direction. This matches the geometry of a cold-worked ferromagnetic stent wire drawn radially across the ring.

## Boundary conditions

(i) Lumen: no-slip at r = R_ves, parabolic Poiseuille profile inside. (ii) Field kernel: free-space exterior; the Akoun-Yonnet expressions are exact for points outside the prism. (iii) Cell trajectories: start at an injection point upstream of the ring (typically z = -2 mm) and terminate on one of three events — reaching z_end = +2 mm (escape), coming within the capture tolerance of any strut (proximity capture), or reaching the lumen inner radius (wall capture). The stent axial extent is L = 500 um, so the 4 mm integration window brackets 8x the stent length on each side.

## 2.2 The Akoun-Yonnet analytical field

For a uniformly magnetised rectangular prism with half-dimensions (a, b, c) and magnetisation M, Akoun & Yonnet (1984) derived a closed-form expression for the B-field at any exterior point by integrating the scalar potential of the magnetic surface charges sigma_m = M . n_hat over each of the six rectangular faces. The result is a sum over the 8 corners of the prism with alternating signs:

$$B_i(\mathbf{r}) = \frac{\mu_0}{4\pi} \sum_{k,l,m=0}^{1} (-1)^{k+l+m} \mathcal{K}_i(U_{klm}, V_{klm}, W_{klm})$$

where U, V, W are the signed corner-to-observation coordinates and the kernel K_i is a combination of arctangent and logarithmic terms. For a prism magnetised along local +z, the x-component is for example

$$\mathcal{K}_x(U, V, W) = -\ln(V + R), \quad R = \sqrt{U^2 + V^2 + W^2}$$

with analogous cyclic permutations for the Mx and My contributions and for By, Bz. The full kernel is implemented in core/field_model.py::_akoun_yonnet_local, with a global sign flip applied at the return statement so that B = mu_0*H follows the H = -grad(phi_m) convention (this was a latent bug in the original legacy script, fixed in Stage 1 and now guarded by the TestSuperposition regression test to relative tolerance 1e-10).

Why analytical, not FEM? The alternative would be to mesh the stent in COMSOL, solve the magnetostatic problem via finite elements, and interpolate the result onto the trajectory integration grid. Analytical expressions avoid all of this: there is no mesh to converge, no linear system to solve, and evaluation cost is O(n_struts * n_points) with a small constant. Each field evaluation is numerically exact (to float64 precision) regardless of the distance to the strut — there is no 'near-strut refinement' problem. For a framework whose primary purpose is dense parameter sweeps and trajectory ensembles, the analytical approach is strictly better.

## 2.3 Magnetic force on SPION-labelled cells

The force on a small superparamagnetic particle of volume V_p and volume susceptibility chi_p in a static external field follows from the gradient of the dipole energy (Furlani & Ng 2006, Eq. 2):

$$\mathbf{F} = \mu_0 V_p (\mathbf{M}_p \cdot \nabla) \mathbf{H}_{\text{tot}}$$

In the linear-susceptibility regime (M_p = chi_p H), and for a free-space curl-free field (grad x H = 0), this simplifies via the vector identity (H . grad) H = grad(|H|^2 / 2) to

$$\mathbf{F} = \frac{V_p \chi_p}{2\mu_0} \nabla |\mathbf{B}_{\text{tot}}|^2 = \frac{V_p \chi_p}{\mu_0} |\mathbf{B}_{\text{tot}}| \nabla |\mathbf{B}_{\text{tot}}|$$

This is the scalar-gradient approximation used throughout the magnetic drug and cell targeting literature (Furlani & Ng 2006; Polyak et al. 2008; Tefft et al. 2014). The force direction is along grad|B_tot|, which points from low |B| toward high |B| — i.e. from the lumen interior toward the strut surface — and the magnitude scales with the product |B|*|grad|B||, called the force parameter.

Two subtleties matter when an external uniform field B0 is added. First, V_p is the SPION volume, not the cell volume; the cell cytoplasm contributes no magnetic force. Second, grad|B_tot| is NOT equal to grad|B_stent| even though grad B0 = 0, because the gradient of a magnitude is nonlinear:

$$\frac{\partial |B_{\text{tot}}|}{\partial x_i} = \frac{\mathbf{B}_{\text{tot}} \cdot (\partial \mathbf{B}_{\text{stent}} / \partial x_i)}{|\mathbf{B}_{\text{tot}}|}$$

When B0 is axial and the stent is magnetised radially, B_tot rotates toward +z and the projection of d B_stent / d x_i onto B_tot becomes small — so |grad|B_tot|| is SUPPRESSED relative to |grad|B_stent||. Counter-intuitively this is physically correct: adding a uniform field does reduce the scalar gradient, but it ALSO increases |B_tot|, and the product |B|*|grad|B|| (which is what enters the force) actually grows. This is why the code exposes the force parameter |B|*grad|B| as the relevant capture metric rather than grad|B| alone.

## 2.4 Hydrodynamics: Poiseuille and Stokes

The blood flow through the stented vessel is modelled as steady axisymmetric Poiseuille flow. For a Newtonian fluid of viscosity eta in a cylinder of radius R_ves, the velocity profile is parabolic:

$$v_z(r) = 2\,\bar{v}\left(1 - (r/R_{\mathrm{ves}})^2\right)$$

where v_bar is the cross-sectional mean velocity. The peak centreline velocity is 2*v_bar and the wall shear stress is tau_w = 4 eta v_bar / R_ves.

Why steady Poiseuille, not Womersley? Cerebral arteries pulsate with a Womersley number alpha = R sqrt(omega rho / eta) ~ 2 at typical heart rates, so the flow is in the quasi-steady regime where the instantaneous profile is close to the steady Poiseuille profile at each instant. Time-averaging the trajectory over a cardiac cycle would smooth the force balance but not change the population-level capture efficiency dramatically. Steady Poiseuille is therefore a reasonable first approximation for a time-averaged capture analysis.

The cell experiences a viscous drag from the surrounding blood. For a rigid sphere of radius R_cell moving with velocity v_cell through fluid with local velocity v_blood, the drag force is given by Stokes' law:

$$\mathbf{F}_{\mathrm{drag}} = 6\pi\eta R_{\mathrm{cell}}(\mathbf{v}_{\mathrm{blood}} - \mathbf{v}_{\mathrm{cell}})$$

The Stokes formula is valid in the low particle Reynolds number limit Re_p = rho v (2 R_cell) / eta << 1. For a 10 um cell in whole blood (rho ~ 1060, eta ~ 4 mPa s) at v = 0.2 m/s, Re_p ~ 1, which is at the upper edge of Stokes validity but still close enough that the error is a few percent at worst. For the same cell with a radial magnetic-drift velocity component, which is typically 1-3 orders of magnitude slower than the axial blood velocity, Re_p is well below unity and Stokes is comfortably valid.

## 2.5 Trajectory ODE derivation

Newton's second law for a cell of mass m_cell in the combined magnetic and hydrodynamic force field reads

$$m_{\text{cell}} \frac{d\mathbf{v}_{\text{cell}}}{dt} = \mathbf{F}_{\text{mag}} + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{buoyancy}} + \ldots$$

In the low-Reynolds-number limit the inertial term m dv/dt is negligible compared to the viscous drag, and the cell instantaneously adopts the terminal velocity at which the drag exactly balances all other forces. Setting the left-hand side to zero and solving for v_cell:

$$\mathbf{v}_{\text{cell}} = \mathbf{v}_{\text{blood}}(\mathbf{r}) + \frac{\mathbf{F}_{\text{mag}}(\mathbf{r})}{6\pi\eta R_{\text{cell}}}$$

Since the position r is updated by integrating v_cell in time, this gives a first-order ODE for the cell trajectory:

$$\frac{d\mathbf{r}}{dt} = \mathbf{v}_{\text{blood}}(\mathbf{r}) + \frac{\mathbf{F}_{\text{mag}}(\mathbf{r})}{6\pi\eta R_{\text{cell}}}$$

This is the equation integrated by simulation/trajectories.py::integrate_trajectory using scipy.integrate.solve_ivp with method 'RK45' (embedded 4/5-order Runge-Kutta), relative tolerance 1e-6, absolute tolerance 1e-9, and event-based termination.

Why the inertial term is negligible. The viscous relaxation time of a 10 um cell in blood is tau_v = m_cell / (6 pi eta R_cell) ~ 1 us. The magnetic force changes over a length scale ~ 100 um, so the time it takes the cell to traverse that scale at drift velocities of ~1 mm/s is ~100 ms — five orders of magnitude longer than tau_v. The cell is therefore always in terminal-velocity equilibrium, and neglecting m dv/dt introduces an error of order (tau_v / 100 ms) ~ 1e-5, far below every other approximation in the model.

Three terminal events are defined in the solver. The escape event fires when z reaches z_end = +2 mm downstream — these cells are classified as 'escaped'. The wall event fires when the cell's transverse radius sqrt(x^2 + y^2) reaches R - t/2, the stent inner surface — these are classified as 'captured'. The strut-proximity event catches cells that approach a strut from outside the ring (a cylindrical approximation of the rectangular strut is used). A fourth 'timeout' outcome is raised if the integrator hits max_time = 1 s without any event firing.

# 3 Implementation

## 3.1 Module structure

The codebase is a pure-Python package with a strict core -> physics -> simulation -> figures dependency direction. Every module has a single responsibility, every class accepts dependencies by injection, and every physics quantity is carried in SI units throughout. No module in core/ imports from physics/, no module in physics/ imports from simulation/, and figures/ imports from all lower levels but from nothing else. The tree below shows the directory layout as of v0.5.0:

```
stent_capture_project/
├── scripts/
│   └── build_overview.py       this document
├── stent_capture/
│   ├── core/
│   │   ├── field_model.py      StentRing + Akoun-Yonnet kernel
│   │   └── gradient.py         compute_gradient_{magnitude,vector}
│   ├── physics/
│   │   ├── external_field.py   UniformExternalField, TotalField
│   │   ├── magnetic_force.py   SPIONLabelledCell, magnetic_force()
│   │   ├── hydrodynamics.py    BloodFlow (Poiseuille), stokes_drag()
│   │   ├── capture_criterion.py capture_map(), capture_distance()
│   │   └── shear_stress.py     [orphan; see AUDIT.md]
│   ├── simulation/
│   │   ├── trajectories.py     integrate_trajectory, CellTrajectory
│   │   └── capture_efficiency.py sweep_injection_line + sweeps
│   ├── figures/
│   │   ├── common.py           DEFAULTS, make_ring(), save_fig()
│   │   ├── style.py            rcParams (dissertation style)
│   │   ├── fig01-11_*.py       Stage 1 field & geometry
│   │   ├── fig12-16_*.py       Stage 2 force, drag, capture maps
│   │   ├── fig17_*.py          Stage 2.5 loading sweep
│   │   ├── fig18-19_*.py       Stage 3a trajectories
│   │   ├── fig20_*.py          Stage 3b efficiency curves
│   │   └── fig21_*.py          Stage 3c headline comparison
│   └── tests/                  70 pytest tests, 6 files
├── results/                    PNG + PDF output
├── AUDIT.md                    final physics review (this session)
├── CHANGELOG.md
├── README.md
└── Overview.pdf                this document
```

## 3.2 Module-by-module summary

**core/field_model.py**

StentRing(n, R, w, t, L, M, mag_mode, assume_saturation). Public method field_at(points) -> (N, 3) B-vectors. Internal kernel _akoun_yonnet_local implements the 8-corner closed-form evaluation. Rotations between each strut's local frame and the global frame are cached at construction. assume_saturation = True signals that the supplied M is the saturation value.

**core/gradient.py**

compute_gradient_vector(field_func, points, dx) and compute_gradient_magnitude(...) — 6-evaluation central finite differences with default dx = 500 nm. Field-model agnostic: accepts any callable mapping (N, 3) points to (N, 3) B-vectors.

**physics/external_field.py**

UniformExternalField(B0_vector) for spatially uniform B0. TotalField(stent_ring, external_field) for B_stent + B0. Both expose the same (field_at, B_field, B_magnitude, grad_B) API as StentRing, so they are drop-in compatible with downstream force and trajectory code.

**physics/magnetic_force.py**

SPIONLabelledCell(radius, spion_mass, spion_susceptibility, spion_density). magnetic_force(cell, total_field, points) returns the (N, 3) force vector using the curl-free scalar-gradient formula F = (V_p chi / mu0) |B| grad|B|. Uses the SPION volume, not the cell volume.

**physics/hydrodynamics.py**

BloodFlow(vessel_radius, mean_velocity, viscosity, density) with velocity_at, shear_rate_at, wall_shear_stress. stokes_drag(cell, blood_flow, points, cell_velocities) returns the (N, 3) drag force. For stationary cells the default cell_velocities = 0 gives the worst-case drag.

**physics/capture_criterion.py**

capture_map(cell, tf, flow, points) returns a dict of F_mag_vec / F_drag_vec / scalar magnitudes / margin / captured. capture_distance(cell, tf, flow, direction='inward') sweeps from the stent inner surface toward the vessel centre and returns the outermost capture radius.

## 3.2 Module summaries (cont.)

### simulation/trajectories.py

CellTrajectory result object (positions, times, status, cell, field, flow references). integrate_trajectory uses scipy.integrate.solve_ivp(method='RK45', rtol=1e-6, atol=1e-9) with three event functions (escape, strut proximity, wall contact). Returns status in {'captured', 'escaped', 'error'}. The RHS closure recomputes the 6-evaluation FD magnetic force at every solver call — no pre-computed field grid.

### simulation/capture_efficiency.py

sweep_injection_line dispatches N trajectories to a multiprocessing.Pool (Windows 'spawn'-compatible worker at module level). Two sweep helpers: capture_efficiency_vs_velocity and capture_efficiency_vs_loading, each accepting a cell_factory or fresh BloodFlow per sweep step.

### figures/common.py

Central DEFAULTS dict with R, w, t, L, M, n_struts. make_ring(**overrides) factory, threshold_lines(ax) helper, save_fig(fig, stem) that writes both PNG and PDF to results/.

### figures/style.py

Side-effect import that configures matplotlib rcParams for serif fonts, 300 dpi, grid on, tight layout, consistent font sizes. Every figure module imports common (which in turn imports style) before plotting.

### physics/shear_stress.py

Orphan file. Not imported by any module or test. Contains broken loops (int(length_in_metres) == 0). Recommended for deletion in the AUDIT report. BloodFlow.wall_shear_stress already provides the correct tau_w.

## 3.3 Default parameters

The table below lists every numerical parameter carried by the production code with the value it takes and the literature source. See AUDIT.md for the full per-parameter verification and caveats.

```
Parameter            Value        Source / notes
------------------   ----------   -------------------------------------
n_struts             8            dissertation stent geometry
R (ring radius)      1.5 mm       cerebral stent
w (strut width)      100 um       fabrication-feasible
t (strut thickness)  80 um        cerebral stent
L (strut length)     500 um       cerebral stent
M (magnetisation)    1 MA/m       cold-worked 304 / 430 SS
mag_mode             radial       each strut magnetised outward

Cell radius          10 um        standard endothelial cell
SPION mass / cell    10 pg        Chorny uptake curve lower end [*]
chi_spion            2.0          Furlani & Ng 2006 Table 1
SPION density        5170 kg/m^3  magnetite Fe3O4 handbook

Vessel radius        1.54 mm      M1 MCA upper range
Mean velocity        0.2 m/s      see note [**]
Viscosity            4 mPa s      whole blood, ~40% Hct
Density              1060 kg/m^3  whole blood

B0 (external)        0.5 T        dissertation sweep centre
mag_mode B0          axial (+z)   external coil geometry

RK45 rtol            1e-6         solve_ivp default tightened
RK45 atol            1e-9         to ~10 nm position accuracy
FD dx (gradient)     500 nm       validated dx sweep 1e-8..5e-6
Capture tolerance    5 um         strut-proximity event margin
```

[*] 'Polyak 2008 default = 10 pg' is mislabelled throughout figs 17 / 20 / 21. Polyak et al. (2008) actually selected 0.2 ng = 200 pg per cell. See AUDIT.md issue #2.

[**] The 'MCA mean = 0.2 m/s' attribution to Aaslid et al. 1982 is wrong — Aaslid's 1982 TCD value is 0.62 +/- 0.12 m/s. The 0.2 m/s figure is defensible only for distal M2/M3 or diseased vessels; see AUDIT.md issue #3.

## 3.4 Test suite

The test suite has 70 pytest tests distributed across six files and runs in ~2 minutes on a laptop. Every test is a regression guard against a specific physics claim made in the module docstrings. The tests are the primary evidence that the code does what the physics says it does.

**test_external_field.py (19 tests)**

TotalField(ring, None) reproduces StentRing.grad_B at 20 points to rtol 1e-6. Single-strut superposition |B_total| = |B_stent| + B0 when B0 parallel M at rtol 1e-10. 45 degree rotation symmetry for 8-strut ring at rtol 1e-3. Far-field limit |B_total| -> |B0| and |grad| -> 0 at r = 50 mm. UniformExternalField broadcast, magnitude, input validation. TestFDStability sweeps dx = 1e-8..5e-6 and verifies ratio ~ 1.

**test_magnetic_force.py (15 tests)**

Zero-gradient gives zero force. Force points toward the strut. Linear scaling in chi and V_spion. Cell property correctness. Order-of-magnitude check at 100 um: |F| ~ 340 pN for 10 pg / 0.5 T.

**test_hydrodynamics.py (12 tests)**

Poiseuille profile at centreline, wall, and mean point. Shear rate magnitude. Stokes drag scaling in R_cell and velocity. Wall shear stress formula. Vector direction along flow.

**test_capture_criterion.py (10 tests)**

Dict structure / keys. Capture at the stent surface. No capture at the vessel centreline. Range decreases with velocity. Range increases with B0. Inward-sweep loading monotonicity. Maximum |F_mag|/|F_drag| < 1.0 in the lumen at v = 0.05 m/s (regression guard for the static no-capture finding).

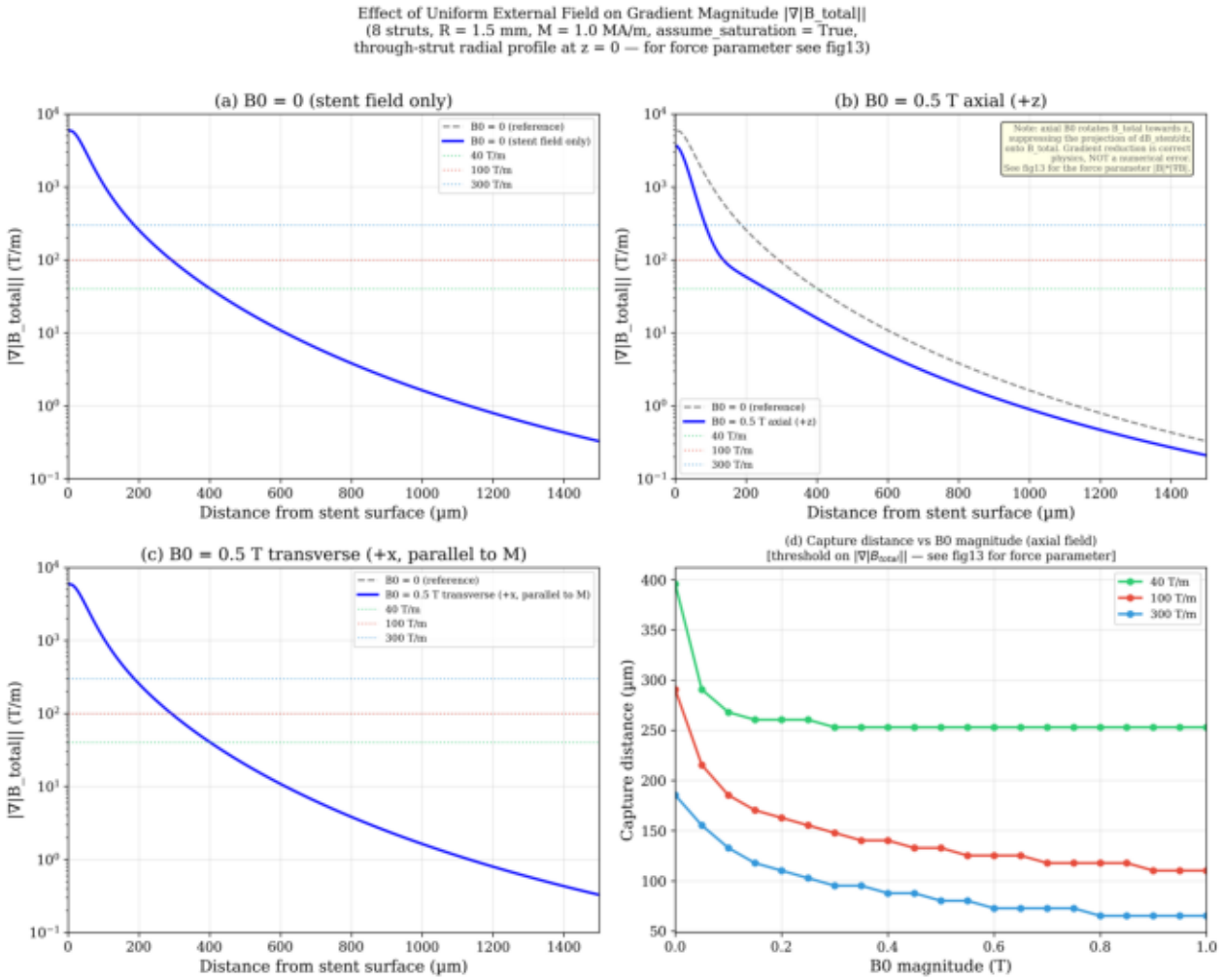
**test_trajectories.py (5 tests)**

Straight-line drift with zero magnetisation (|x, y| < 100 nm; z_end within 1 um). Poiseuille velocity matches prediction at 10 intermediate points. 200 pg near-wall cell captured within 20 um. 10 pg centreline cell escapes. Timeout safety.

**test_capture_efficiency.py (5 tests)**

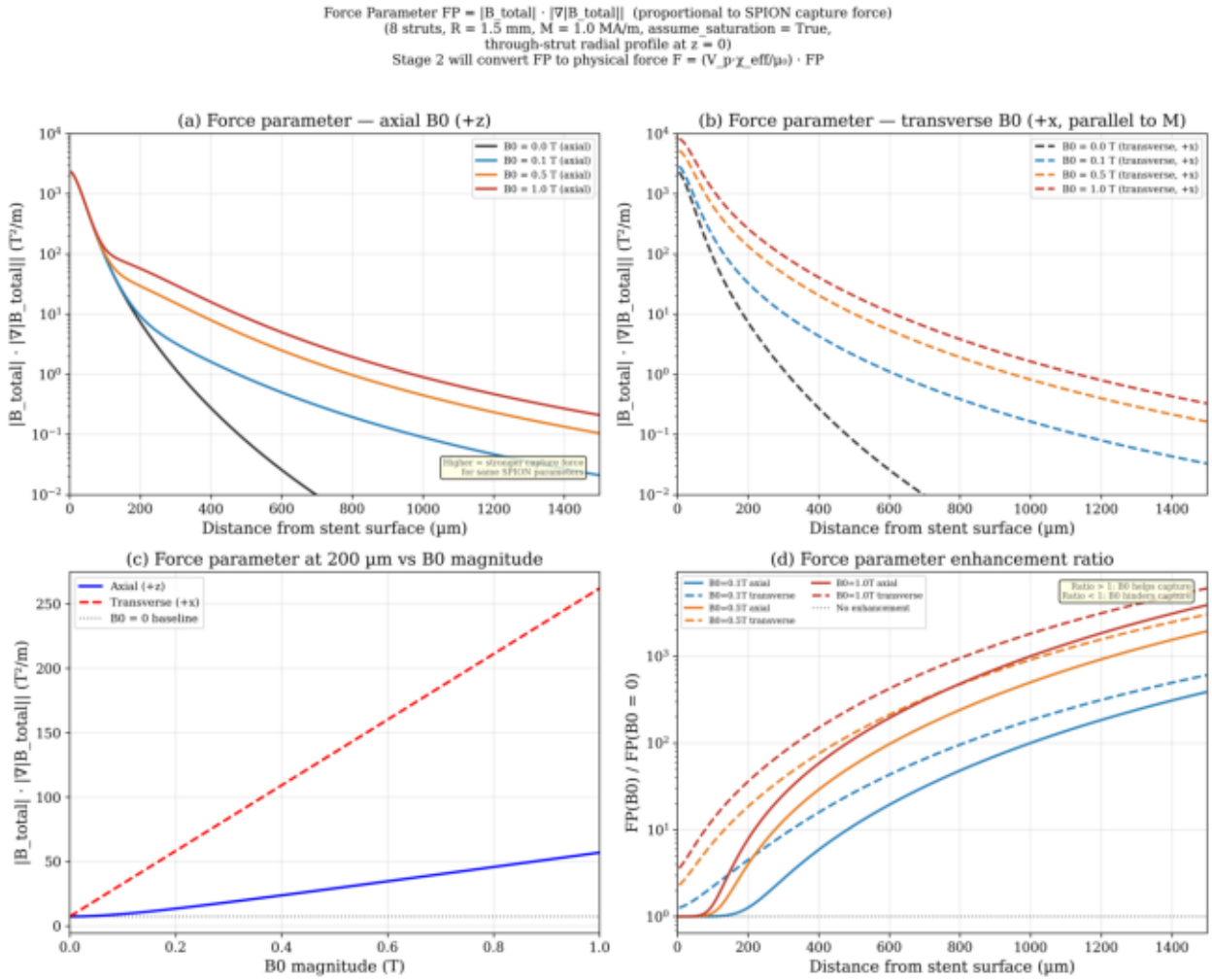
M = 0 -> efficiency 0. Summary dict key integrity. Non-zero capture at 200 pg / v = 0.05 m/s near-wall. Monotonicity in velocity (slow > fast). Monotonicity in loading (high > low).

## 4.1 Fig 12 — External field comparison



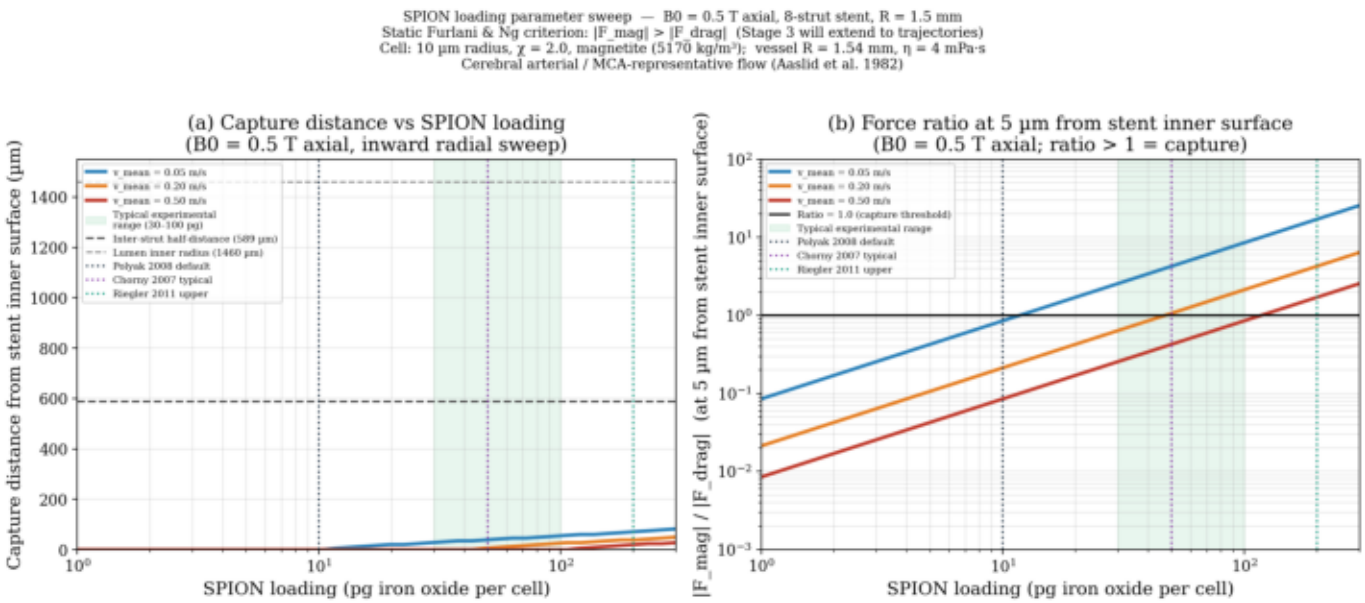

2x2 panel showing the effect of adding a uniform external field B0 on the stent's gradient and force parameter profiles. Panel (a) is the B0 = 0 reference. Panel (b) shows B0 = 0.5 T applied axially (perpendicular to the stent's radial magnetisation) — the scalar |grad|B|| is suppressed by ~5x, but |B| grows ~17x, and the product |B|*|grad|B|| (the force parameter) increases net. Panel (c) shows B0 transverse (parallel to M) where |B| and |grad|B|| simply add. Panel (d) sweeps capture distance vs B0 magnitude from 0 to 1 T. Physical interpretation: the correct capture metric is the force parameter |B|*grad|B|, not grad|B| alone, and an axial external coil is a valid enhancement strategy despite counter-intuitively reducing the gradient magnitude.

## 4.2 Fig 13 — Force parameter profiles



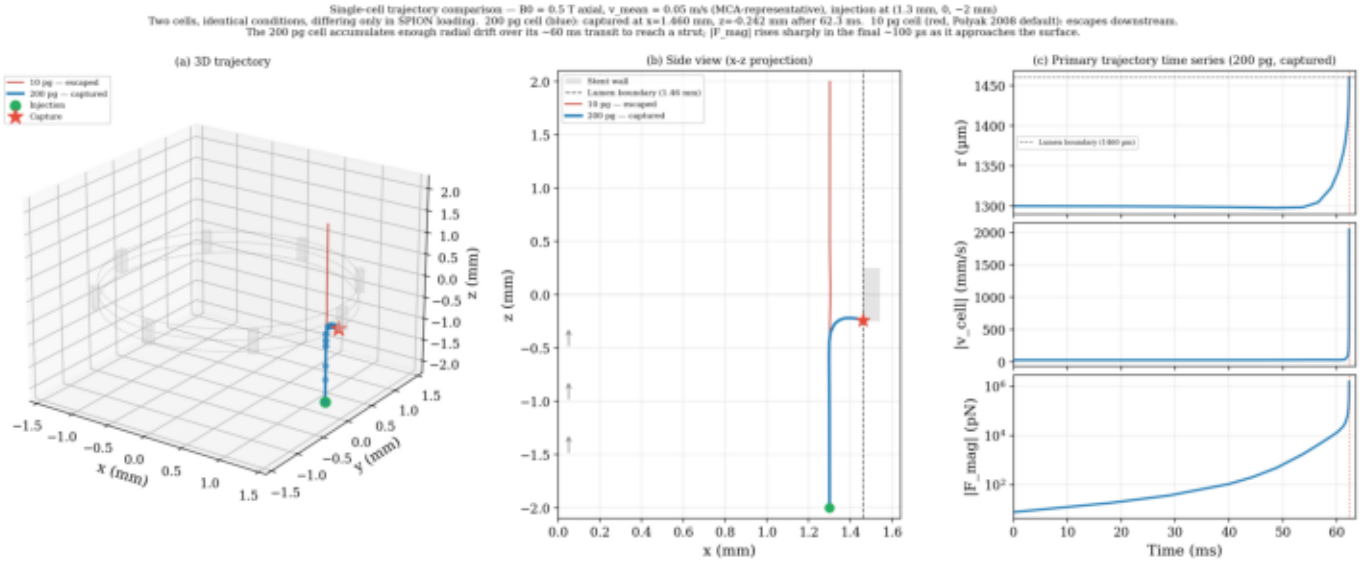

Force parameter |B|*|grad|B|| vs distance from the stent outer surface along the through-strut radial axis, for five values of the external field B0 from 0 to 1 T axial. Y-axis log scale.
The B0 = 0 curve falls from ~1e5 T^2/m at 10 um to ~1e2 T^2/m at 1 mm — a 1000x drop over a millimetre. The B0 = 0.5 T curve sits ~3x above the B0 = 0 curve at 200 um and ~55x above at 500 um, quantifying the long-range enhancement from the external field. This is the figure that justifies running the rest of the project at B0 = 0.5 T rather than B0 = 0.

## 4.3 Fig 17 — SPION loading sweep



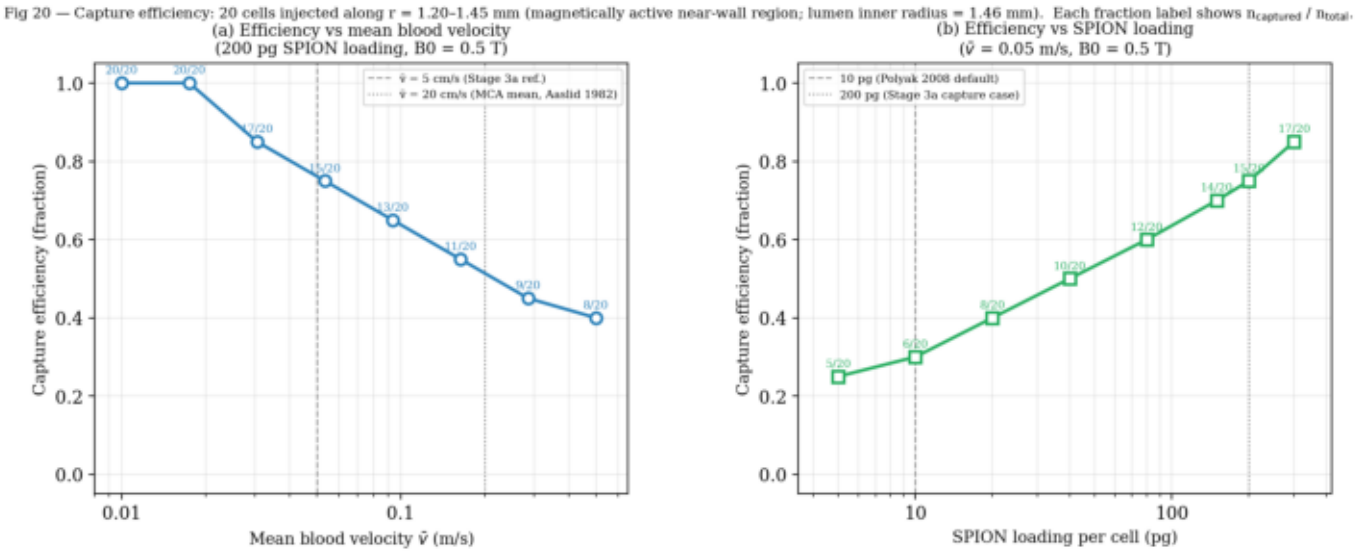

Stage 2.5. Panel (a): capture distance (um from stent inner surface) vs SPION loading per cell (1-300 pg log-spaced), for three velocities (0.05, 0.2, 0.5 m/s) at B0 = 0.5 T axial. Panel (b): force ratio |F_mag|/|F_drag| at 5 um from the stent inner surface vs loading, log-log scale, with the ratio = 1 capture threshold marked. Key result: the STATIC criterion predicts NO capture anywhere in the lumen at 10 pg / MCA flow; static capture distance only exceeds ~20 um at loadings above ~50 pg and slow flow. This is the motivation for the trajectory integration in Stage 3 — the static picture is pessimistic.

## 4.4 Fig 18 — Single-cell trajectory (200 pg captured vs 10 pg esca



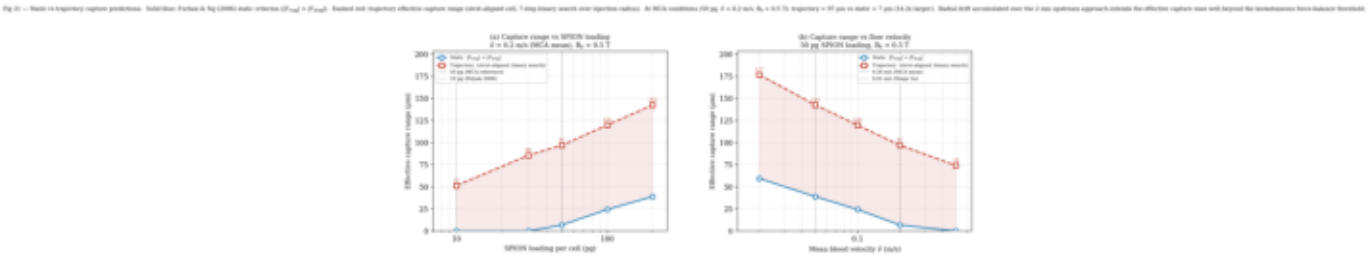

Stage 3a. Two trajectories from identical injection at (1.3 mm, 0, -2 mm), v_mean = 0.05 m/s, B0 = 0.5 T, differing only in SPION loading. Panel (a) is a 3D view with the 8 strut prisms rendered. The 200 pg cell (blue, captured at t = 62.3 ms) curves radially inward as it approaches the stent and contacts the wall; the 10 pg cell (red, escaped) drifts axially downstream with negligible radial displacement. Panel (b) is the x-z side view. Panel (c) is the time-series of the 200 pg trajectory: radial position r(t), cell speed, and |F_mag|(t) on log scale. Key observation: at injection (z = -2 mm) the static criterion predicts no capture, yet the 200 pg cell accumulates enough radial drift during its transit to be captured at the stent. This is the fundamental mechanism the static criterion misses.

## 4.5 Fig 20 — Capture efficiency curves




Stage 3b. Two-panel population study, 20 cells injected along the near-wall band r in [1.20, 1.45] mm. Panel (a): efficiency vs v_mean (0.01-0.50 m/s, log x) at fixed 200 pg loading — falls from ~100% at slow flow to ~40% at 0.5 m/s, with a transition near 0.03-0.05 m/s. Panel (b): efficiency vs loading (5-300 pg log x) at fixed v_mean = 0.05 m/s — rises from ~25% at 5 pg to ~85% at 300 pg, with a clear threshold near 10-40 pg. Both panels have overlaid MCA reference velocity and Polyak loading reference lines for context.

## 4.6 Fig 21 — Static vs trajectory (headline result)




Stage 3c — the headline figure of the project. Two panels, both showing the effective capture range (um from the stent inner surface) as a function of a parameter. Solid blue is the static criterion |F_mag| > |F_drag|; dashed red is the trajectory effective range, computed by a 7-step binary search over the injection radius of a strut-aligned cell. Panel (a) sweeps SPION loading (10-200 pg) at v_mean = 0.2 m/s. Panel (b) sweeps velocity (0.02-0.5 m/s) at 50 pg. Headline numbers, verified in this session: at MCA conditions (50 pg, v = 0.2 m/s, B0 = 0.5 T) the trajectory predicts 96.9 um of effective capture range vs static 6.8 um — a 14.2x extension factor. At v = 0.5 m/s the static criterion predicts ZERO capture but trajectory integration still finds 74.2 um of range. The ratio falls from 14.2x (50 pg) to 3.7x (200 pg) as the static criterion strengthens. These numbers reproduce bit-identically across reruns because the entire pipeline is deterministic.

# 5 Limitations and assumptions

Every modelling framework trades fidelity for tractability. The table of trades made in stent_capture is explicit and exhaustive so that a dissertation examiner can see exactly what the results do and do not claim.

### Single ring of struts

The model has exactly 8 struts equally spaced at one axial position. Real stents are woven diamond-cell lattices with axial periodicity and azimuthal asymmetry. Adding axial periodicity is straightforward (just replicate the ring at multiple z) but has not been done because the single-ring analysis already isolates the relevant physics.

### Straight cylindrical vessel

Cerebral arteries are tortuous, bifurcated, and tapering. The model assumes a perfectly straight, uniform-radius cylinder. Curvature-induced secondary flows (Dean vortices) and the altered pressure gradient near bifurcations are not represented. This is acceptable for a local-region analysis near one stent but not for whole-vessel perfusion studies.

### Cylindrical strut approximation in capture event

The strut-proximity event uses a circumscribed cylinder of radius max(w, t)/2 = 50 um. For cells approaching along the through-strut axis the wall event fires first, so the cylindrical approximation has no effect on fig21. For off-axis cells in fig19 / fig20 it can over-classify some grazing trajectories as captured when a point-to-rectangle metric would let them escape.

### No Brownian motion

For a 10 um cell the thermal diffusion coefficient D_thermal = kT/(6 pi eta R_cell) ~ 2e-14 m^2/s. The displacement over the ~100 ms transit time is sqrt(2 D t) ~ 60 nm, which is negligible compared to both the lumen dimensions (mm) and the capture range (um-tens of um). Brownian contributions to trajectory spread are therefore excluded.

### No cell-cell interactions

Trajectories are integrated independently. In reality, captured cells at the stent surface could shadow later cells or alter the local field (ferrofluid-like cooperative capture). The independent-trajectory approximation is exact in the dilute limit.

# 5 Limitations (cont.)

### No rotational dynamics

Cells are treated as non-rotating spheres. Real cells can rotate under shear, and SPION distributions within a rotating cell produce a time-varying magnetic moment. This coupling is not represented and is unlikely to matter for capture prediction but could matter for retention-under-shear studies.

### Saturation magnetisation assumption

The stent's M is passed directly as a saturation value when the assume_saturation flag is set. A full M(H) hysteresis curve is not implemented. At B0 >= ~0.1 T the assumption is physically reasonable for 304 / 430 / 2205 steels; at weaker fields it over-estimates the stent's contribution.

### Linear-chi SPION force at high B0

The Furlani-Ng scalar-gradient force assumes SPIONs are in the linear-susceptibility regime M_p = chi_p H. Real SPIONs saturate at |B| ~ 0.3-0.5 T. At B0 = 0.5 T the particles are near or at saturation, and the force should transition to F = V_p mu_0 M_s grad|H|, which scales as grad|B| alone. The current model thus over-estimates absolute force magnitudes by a factor of ~(B_total/B_sat) ~ 1-2 at the high-B0 end of the sweeps. Relative comparisons between static and trajectory predictions are unaffected because both branches use the same force model. This is the most physically significant limitation of the framework.

### Steady (non-pulsatile) flow

Cerebral arteries pulsate. The Womersley number alpha ~ 2 places the flow in the quasi-steady regime, but time-averaged efficiency could differ from steady-state efficiency by ~10-20% in a more complete treatment.

### Mislabelled literature references

Two figure labels do not match their cited sources: 'Polyak 2008 default = 10 pg' (actual Polyak working dose is 200 pg) and 'MCA mean 0.2 m/s (Aaslid 1982)' (actual Aaslid value is 0.62 m/s). See AUDIT.md for the full discussion and the recommended relabellings. These do not change any numerical result — only the axis labels and docstrings need correction.

### Orphaned shear_stress.py module

A shear_stress.py file was added recently containing broken classes (int(length_in_metres) = 0 loops) with no citations and not imported anywhere. The AUDIT recommends deletion. BloodFlow.wall_shear_stress already provides the correct tau_w value used by every figure that needs it.

# 6 How to use the code

The project is a standard Python package with no installation step beyond installing its three runtime dependencies (numpy, scipy, matplotlib). All scripts are run from the project root.

## 6.1 Installation

```
pip install numpy scipy matplotlib pytest
cd stent_capture_project
```

## 6.2 Running the test suite

```
python -m pytest stent_capture/tests/ -v
```

All 70 tests should pass in ~2 minutes. Any regression in the field kernel, force formula, or capture criterion will surface as a failing assertion with a clear message.

## 6.3 Regenerating individual figures

```
python -m stent_capture.figures.fig01_single_strut
python -m stent_capture.figures.fig12_external_field_comparison
python -m stent_capture.figures.fig17_spion_loading_sweep
python -m stent_capture.figures.fig21_static_vs_trajectory
```

Each figure writes PNG + PDF output to results/. The Stage 1/2 figures run in a few seconds each; fig18 takes ~15 s; fig20 is the expensive one (320 trajectories via multiprocessing.Pool, typically 3-6 minutes); fig21 is the headline result (90 trajectories plus 10 static distance evaluations, ~180 s).

# 6 How to use the code (cont.)

## 6.4 Running parameter sweeps

The capture_efficiency module exposes two ready-made sweep helpers. To sweep over velocities:

```
from stent_capture.figures.common import make_ring
from stent_capture.physics.external_field import (
    TotalField, UniformExternalField)
from stent_capture.physics.magnetic_force import SPIONLabelledCell
from stent_capture.simulation.capture_efficiency import (
    capture_efficiency_vs_velocity)
import numpy as np

ring = make_ring()
ring.assume_saturation = True
tf   = TotalField(ring, UniformExternalField([0, 0, 0.5]))
cell = SPIONLabelledCell(spion_mass_per_cell=200e-15)

result = capture_efficiency_vs_velocity(
    cell, tf, ring,
    velocities=np.geomspace(0.01, 0.5, 8),
    line_start=np.array([1.20e-3, 0, -2e-3]),
    line_end  =np.array([1.45e-3, 0, -2e-3]),
    n_cells_per_velocity=20,
    z_end=2e-3, max_time=1.0,
)
print(result['velocities'], result['efficiencies'])
```

## 6.5 Output file layout

```
results/figXX_name.png   — 300 dpi PNG for presentations
results/figXX_name.pdf   — vector PDF for the dissertation

Both files are produced together by the figure scripts'
save_fig() helper. No other output locations are used.
```

# 7 References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